

Galactic Structure, Galaxies & the ISM

Problem Set

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This problem set accompanies the *Galactic Structure, Galaxies & the ISM* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Use $G = 6.674 \times 10^{-11} \text{ N m}^2\text{kg}^{-2}$, $c = 3 \times 10^8 \text{ m/s}$, $M_\odot = 1.989 \times 10^{30} \text{ kg}$, $R_\odot = 6.96 \times 10^5 \text{ km}$, $L_\odot = 3.828 \times 10^{26} \text{ W}$, $1 \text{ AU} = 1.496 \times 10^8 \text{ km}$, and $1 \text{ kpc} = 3.086 \times 10^{19} \text{ m}$ where needed, unless a question tells you otherwise.

Section A — Objective

O1. In an eclipsing binary, the relative transverse speed is $v = 50 \text{ km/s}$, with $(t_4 - t_1) = 6.0 \text{ hr}$ and $(t_3 - t_2) = 2.0 \text{ hr}$. The larger star's radius R_l is closest to:

- (a) $3.6 \times 10^5 \text{ km}$
- (b) $1.8 \times 10^5 \text{ km}$
- (c) $7.2 \times 10^5 \text{ km}$
- (d) $4.2 \times 10^5 \text{ km}$

O2. A star is observed to be both fainter *and* redder than its known spectral type predicts, with no unusual proper motion. This is best explained by:

- (a) the star being farther away than assumed
- (b) interstellar extinction along the line of sight
- (c) the star being more metal-poor than assumed
- (d) the star being an unresolved spectroscopic binary

O3. Using $\kappa = 0.034 \text{ m}^2/\text{kg}$ (electron-scattering opacity), the Eddington luminosity of a $10 M_\odot$ compact object is closest to:

- (a) $1.5 \times 10^{32} \text{ W}$
- (b) $1.5 \times 10^{28} \text{ W}$
- (c) $1.5 \times 10^{36} \text{ W}$
- (d) $5.0 \times 10^{30} \text{ W}$

O4. Which one of the following statements is **false**?

- (a) $A - B$ gives the Sun's local angular orbital velocity, Ω_0 .
- (b) $A + B$ gives the local shear of the rotation curve, $-(d\Theta/dR)_0$.
- (c) $A \neq 0$ confirms the Galaxy does not rotate as a rigid body.
- (d) Seyferts, quasars, and blazars are physically distinct classes of AGN, each powered by a different kind of central engine.

O5. A galaxy's rotation curve is flat at $\Theta = 220$ km/s out to $r = 15$ kpc. The enclosed mass $M(r)$ is closest to:

- (a) $1.7 \times 10^{11} M_\odot$
- (b) $1.7 \times 10^9 M_\odot$
- (c) $1.7 \times 10^{13} M_\odot$
- (d) $3.4 \times 10^{11} M_\odot$

Section B — Subjective

S1. A double-lined, eclipsing spectroscopic binary has period $P = 5.0$ days, radial-velocity amplitudes $K_1 = 80$ km/s and $K_2 = 120$ km/s, and (since it eclipses) inclination $i \approx 90^\circ$. Its light curve gives contact-time differences $(t_4 - t_1) = 10.0$ hr and $(t_3 - t_2) = 3.0$ hr.

- (i) Find the mass ratio M_1/M_2 .
- (ii) Using the mass function $f(M_1) = M_2^3 \sin^3 i / (M_1 + M_2)^2 = PK_1^3 / (2\pi G)$ together with your mass ratio from (i), solve for M_1 and M_2 individually (in $M_\odot$).
- (iii) Using Kepler's Third Law, find the orbital separation a , and hence the relative transverse speed $v = 2\pi a / P$. (You should find $v \approx K_1 + K_2$ — convince yourself briefly why this must be true for an edge-on, circular orbit.) Finally, use v and the contact-time data to find R_l and R_s , in solar radii.

S2. The general Oort constants are $A = \frac{1}{2} \left(\frac{\Theta_0}{R_0} - \left(\frac{d\Theta}{dR} \right)_0 \right)$, $B = -\frac{1}{2} \left(\frac{\Theta_0}{R_0} + \left(\frac{d\Theta}{dR} \right)_0 \right)$.

- (i) Suppose the local rotation curve were exactly flat, $\Theta(R) = \Theta_0 = \text{const}$, near the Sun. Find A and B individually in terms of Θ_0 and R_0 alone, and show that this special case requires $A + B = 0$ exactly.
- (ii) The measured local values are $A \approx 15$ km/s/kpc and $B \approx -13$ km/s/kpc. Using these, find $(d\Theta/dR)_0$, and comment on how close the Milky Way's rotation curve actually is to being locally flat near the Sun.

- (iii) Without redoing the full derivation, explain briefly why $A - B$ (rather than, say, A or B individually) is the combination that isolates $\Omega_0 = \Theta_0/R_0$ cleanly — i.e. what has to cancel, and why the $(d\Theta/dR)_0$ term drops out specifically under subtraction rather than addition.

S3. A quasar has observed luminosity $L = 10^{39}$ W and is inferred to be radiating at exactly its Eddington limit. Take $\kappa = 0.034 \text{ m}^2/\text{kg}$.

- (i) Estimate the mass M of the central supermassive black hole, in solar masses.
- (ii) If this luminosity is entirely powered by accretion at efficiency $\eta \approx 0.1$, find the mass accretion rate $\dot{M}$ required, in both kg/s and $M_\odot/\text{yr}$.
- (iii) Using M from (i) and the Schwarzschild radius formula (Handout 5), estimate R_s for this black hole, convert it to AU, and comment on what your answer implies about the physical size of even a supermassive black hole's event horizon.